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18 February 1975
Superseding
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15 June 1964

MILITARY SPECIFICATION

TEST SET, INDICATOR, AN/ARM-155

This specification has been approved by the Naval Air System Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers one type of equipment designated as Test Set, Indicator, AN/ARM-155. This equipment shall provide input power control, interconnecting cables, channel selectors, indicators and meters for use in intermediate level maintenance of Navy airborne TACAN equipment. It is intended to support the AN/ARM-21, AN/ARN-52, AN/ARN-84, AN/ARN-86, AN/ARN-105, AN/ASQ-19, AN/ASQ-56, AN/ASQ-57, AN/ASQ-58, AN/ASQ-88, TACAN Station-keeping TACAN data link, Inverse TACAN and equivalent airborne TACAN systems (see paragraph 6.1 Intended Use.)

1.2 Classification. The Test Set, Indicator, AN/ARM-155 shall consist of the following items:

Control, Indicator, C-8963/ARM-155	3.5.1
Transit Case, Control, Indicator Test Set, AN/ARM-155	3.5.2
Indicator, Control, ID-1812/ARM-155	3.5.3
Transit Case, Indicator, Control, Test Set, AN/ARM-155	3.5.4
Mounting Base, MT-4410/ARM-155	3.5.5
Transit Case, Mounting Base, Electrical Equipment, AN/ARM-155	3.5.6
Cables and Adapters	3.5.7.1

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1.3 Associated equipment. The Test Set, Indicator, AN/ARM-155 shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-5824	Course Indicator ID-250A/ARN
MIL-R-7060	Radio Set AN/ARN-21(*)
MIL-M-7793	Meter, Time Totalizing
MIL-I-8303	Indicator, Course ID-351/ARN
MIL-N-8689	Network, Phase Detecting CV-279/ARN
MIL-I-8691	Indicator, Azimuth, ID-307(*)/ARN
MIL-I-8692	Indicator, Range, ID-310(*)/ARN
MIL-I-8693	Indicator, Coupler CU-395/ARN-41A
MIL-I-9229	Indicator, Course ID-387/ARN
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Electromechanical Equipment and Associated Repair Parts
MIL-T-18300	Design Data for Avionic Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection for Airborne Electronic Equipment, Format for

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Military (Continued)

MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-I-18955	Indicator, Range ID-388/ARN
MIL-S-20708	Synchros, General Specification for
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-I-22075	Indicator, Bearing-Distance-Heading ID-663/U
MIL-N-22239	Navigational Set, TACAN AN/ARN-52(V)
MIL-R-22677	Receiver-Transmitter, Radio RT-541(*)/ASQ and Decoder, Pulse KY-309(*)/ASQ
MIL-R-22763	Receiver-Transmitter, Radio RT-547(*)/ASQ-19 and Decoder, Pulse KY-312(*)/ASW-19
MIL-I-23366	Indicator, Horizontal Situation ID-1013(*)/a
MIL-H-27848	Horizontal Situation Indicator AQU-4/A
MIL-R-81105	Receiver-Transmitter, Radio RT-736(*)/ASQ-88 and Decoder, Pulse KY-531(*)/ASQ-88
MIL-D-81170	Data Link Set, AN/ASW-27
MIL-N-81207	Navigational Set AN/ARN-84

STANDARDS

Military

MIL-STD-108	Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment
MIL-STD-189	Racks, Electrical Equipment, 19-inch and Associated Panels
MIL-STD-200	Electron Tubes, Selection of

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Military (Continued)

MIL-STD-287	Test Signals for Electronic Tactical Air Navigation Equipment (TACAN)
MIL-STD-291	Standard Tactical Air Navigation (TACAN) Signal
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definition and System of Units, Electromagnetic Interference Technology
MIL-STD-470	Maintainability Program Requirements
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-721	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors and Safety
MIL-STD-781	Reliability Test: Exponential Distribution
MS 21993	Cable Assemblies, Power, Electrical

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionic Equipment and Systems, General Requirements for
WR-97	Aircraft Electronics Equipment, Reliability Requirements for Design of

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2.1.1 Availability of documents.

(When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Parts and material. In the selection of parts and material fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

1. Microelectronic items shall conform to the requirements specified herein.

2. Other parts and material shall conform to MIL-T-21200.

3. Nonrepairable subassemblies, as outlined in MIL-T-21200, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (See 3.3.7 and 6.3).

4. When previously produced models of this equipment did not use nonrepairable assemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

5. When microelectronic modular assemblies are used, such assemblies shall meet the requirements of AR-5.

3.2.1 Non-standard parts and material approval. Approval for the use of nonstandard parts and material shall be obtained as outlined in MIL-T-21200.

3.2.2 Electron devices. Transistors and diodes shall be chosen and applied as outlined in MIL-T-21200.

3.3 Design and construction. This equipment shall conform to the applicable requirements of MIL-T-21200 for design and construction unless otherwise specified herein.

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3.3.1 Total weight. The total weight of the equipment shall not exceed 408 pounds including cable assemblies and case.

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with Requirement 35 of MIL-STD-454.

3.3.2.1 Operational stability. The equipment shall operate with satisfactory performance for 1440 hours, continuously or intermittently without necessity for readjustment of any controls which are not accessible from the control panel. The equipment shall meet the stability requirements under both the following conditions.

1. Unstacked configuration (i.e.), side by side with associated equipment, Test Set, Radio, AN/ARM-156.

2. Stacked configuration (i.e.), in a vertical column with associated equipment, Test Set, Radio, AN/ARM-156.

3.3.2.2 Operating life. The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be determined by the manufacturer and submitted to the procuring agency.

3.3.2.3 Mean-time-between-failures (MTBF). The equipment, including any built-in test provisions, shall have a minimum acceptable (operating) MTBF (θ_1) of 750 hours when tested as specified in 4.4.3. The equipment's inherent design MTBF shall be 1500 hours. The six electro-mechanical instruments incorporated in the Power Control Unit by NAVAIR direction, are not included in the MTBF requirement. The instruments so excluded are:

Course Indicator ID-387
Bearing-Distance Indicator-Bendix ID-663 (Three required)
Master Range Indicator-Bendix 7909-1L9-A1 or equivalent
Master Bearing Indicator-Bendix 7003-1B-2-A1 or equivalent

3.3.2.4 Time totalizing meter. Units 1 and 2 of this equipment shall contain time totalizing meters in accordance with MIL-M-7793. The meters shall be mounted on the front panels in a position readily discernable to the operator.

3.3.3 Cabling and connections.

3.3.3.1 Cables and connectors. The equipment shall provide for the use of cables and connectors in accordance with the requirements of MIL-T-21200 except as noted in Table I of this specification. Insofar as practical, all control panel connectors shall be mounted along the lower edge of the control panel and in no case shall their location be such that the connecting cables will interfere with the operator.

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TABLE I

LIST OF CONNECTORS ON FRONT PANEL OF CONTROL INDICATOR, C-8963/ARM-155

ITEM	FUNCTION	AN/ARM-31 SYMBOL	AN/ARM-155 SYMBOL	NOTES
1	ID-387/ARN	J901	J901*	Used with shorting plug to connect internal indicator.
2	Ident Tone	J902	J902	
3	ID-250A/ARN	J903	J903*	Used with shorting plug to connect internal indicator.
4	115V, 380-420 Hz	J905	J905	
5	28 VDC	J906	J906	
6	ID-388/ARN	J907	J907	Used with shorting plug to connect internal indicator.
7	ID-307/ARN	J908	J908*	
8	ID-310/ARN	J909	J909*	
9	CV-279/ARN-21	J910	J910*	
10	CU-395/ARN-21	J911	J911*	
11	AN/ARN-21	J912	J912*	
12	AN/ARN-21	J913	J913*	
	Control Unit	J922	J922	Connector located in AN/ARN-21 control unit adapter
13	Composite Control Unit	None	J922/J925	Mounted in control unit recess
14	AN/ARN-84	None	J923*	Connect test set to test fixture
15	T.S. (Unit to Unit)	None	J924	Interconnect power and signals from the Power Control and Instrument Unit to the Display Unit of this test set.

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* Connectors whose symbols are accompanied by an asterisk (*) shall be Amphenol 26 Series to provide compatibility of this test set with the cables and test bench harnesses utilized with Test Set, Indicator, AN/ARM-31.

3.3.3.2 Interconnecting cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except as necessary to meet the interference control requirements.

3.3.3.3 Cable assemblies, power electrical. Applicable input power cables and connections shall conform to MS 21993. Adapters shall be provided for ship and shore shop uses.

3.3.3.4 Compatibility with existing cables. The test set shall be fully compatible with any cable or adapter now used with the AN/ARM-31 Test Set, Indicator.

3.3.4 Mechanical requirements.

3.3.4.1 Control panel. All operational controls, connectors, adjustments and indicators shall be located on the control panel and shall conform to MIL-STD-454, Requirement 28. The control panels of 3.5.1.1 and 3.5.3.1 shall be standard 19-inch relay rack panels fastened to an instrument case by captive screws and shall conform to the requirements of MIL-STD-189. The panels with associated equipment thereon shall be removable from the instrument cases for installation in a standard 19-inch relay rack. The controls, indicators, connectors, etc., mounted on the control panels shall be protected by two guardrail type handles mounted one on either side of the panel. These handles shall be adequate for use in carrying the equipment and no other handles shall be provided.

3.3.4.2 Instrument case. The instrument cases shall be constructed in accordance with MIL-T-21200 and shall be shaped to stack above or below and clamp together to other instrument cases to which the instrument may normally stack. The instrument case, except the control panel thereof, shall be drip-proof in accordance with MIL-STD-108.

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements as defined in MIL-T-21200.

3.3.6 Radio interference control. The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits specified by MIL-STD-461, 462 and 463, except that the portions of the Control Indicator, C-8963/ARM-155, which retain the design of the Test Set, Indicator, AN/ARM-31 shall be required to perform to the same limits as demonstrated by the AN/ARM-31. Requirements for CE03 broadband conducted emissions on the AC power leads shall be relaxed 8 dB for frequencies from 35 to 45 kHz.

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3.3.7 Maintainability. Maintainability shall be in accordance with MIL-T-21200 and AR-10, to the extent specified herein.

1. Modularization. The equipment shall be modularized at all levels of maintenance. The weight and mean-time-between-failure of the equipment units shall meet the requirements of 3.3.1 and 3.3.2.3 respectively. Circuits shall be packaged into discrete replaceable modules of such cost and reliability that disposal-on-failure rather than module repair is the most cost effective logistics support action. The criteria for disposal-on-failures shall be as specified in Figure 1.

2. Standardization. The number of different types of critical parts shall be kept to a minimum in accordance with MIL-T-21200 and AR-10.

3. Encapsulation and conformal coatings. Modules or assemblies shall meet the requirements of AR-10, see 3.1.5.

4. Built-In Test (BIT) function. BIT features shall be as specified in 3.5.3.4.4.

5. Test points. Test points shall be provided for all levels of maintenance and shall provide safe and convenient access to electronic circuits. The test points provided shall enable the operator to determine the performance level of the equipment unit and in the event of a failure, to ascertain which replaceable assembly contained therein is at fault.

6. Shop fixed interface. Replaceable assemblies shall not require adjustment or trimming upon installation into an equipment unit with the exception of the Shop Replaceable Assemblies (SRA's) of the power supplies.

7. Quick replaceable assembly. Replaceable assemblies shall, to the maximum extent possible, be designed as quick Replaceable Assemblies as defined in AR-10, see 6.1.5.

8. Equipment repair time (ERT). The equipment shall meet the following:

   a. Corrective maintenance shall not exceed an ERT of one hour when tested in accordance with Method (3) of MIL-STD-471.

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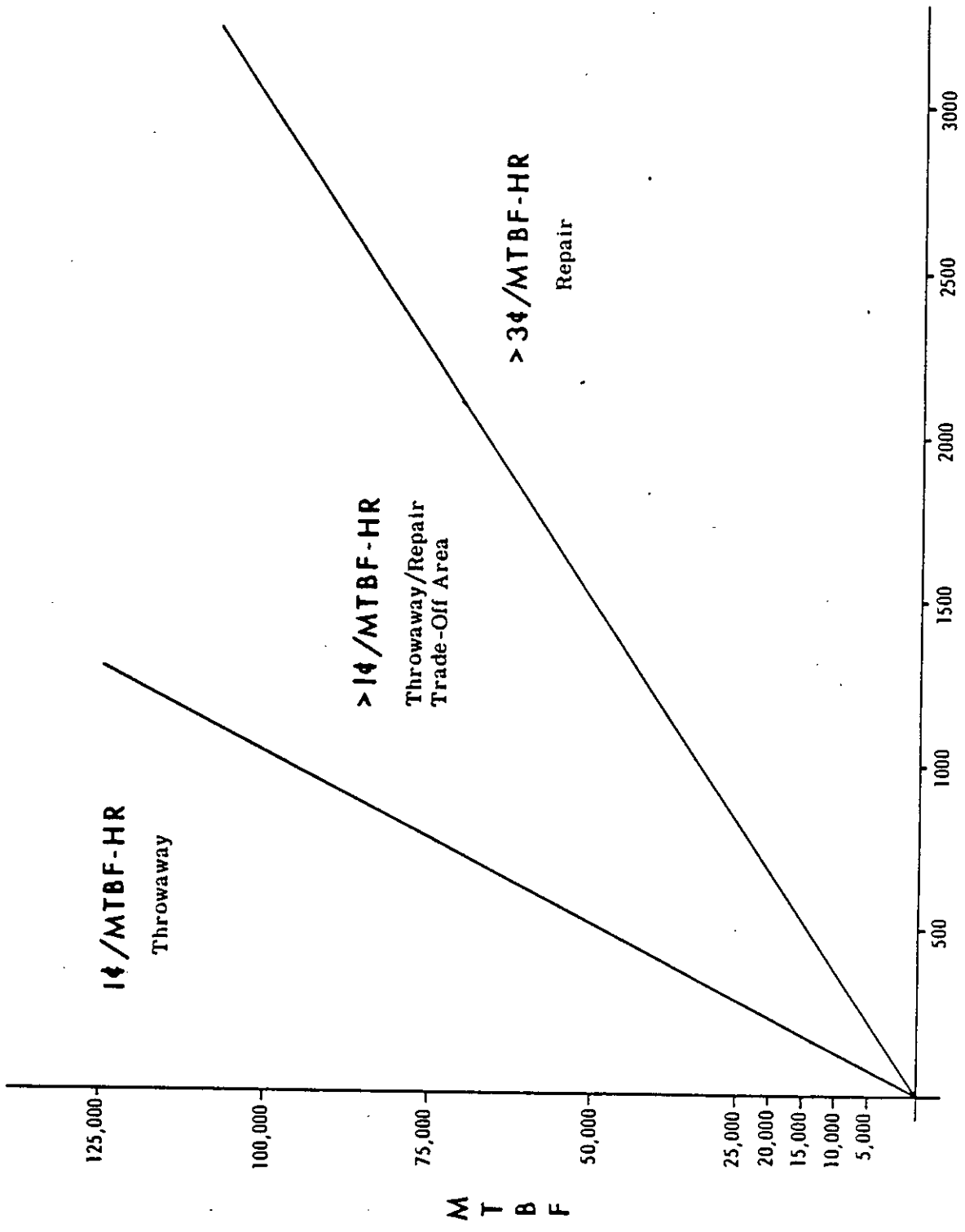


Figure 1. Repair vs Throwaway Criteria

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b. Preventive maintenance time exclusive of calibration cycles shall not exceed one hour for all preventive maintenance actions per maintenance cycle.

3.3.8 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of MIL-N-18307.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ±10° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Voltage	115 ±1V, 400 Hz ±5 Hz, single phase and 27.5 ±0.5 VDC

3.3.10 Service conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combinations of these conditions as specified in MIL-T-21200 for Class 3 equipment, except that the lowest operating temperature shall be -20° C.

3.3.11 Warm-up time. After a warm-up time of 5 minutes under standard conditions and 15 minutes under service conditions, performance specified shall be obtained and maintained for at least 1440 hours continuously without adjustment of any controls not available on the front panel.

3.3.12 Input electrical power. The equipment shall meet applicable requirements of MIL-T-21200 and shall give specified performance from the following power sources with characteristics as defined in MIL-T-21200 having limits specified herein except that operation from a 60 Hz power source shall not be required:

1. Single-phase alternating current power within a frequency range of 380 to 420 Hz and voltages between 103.5 and 126.5 volts rms.

2. Direct current power within a voltage range of 24 to 29 volts.

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3.3.13. Voltage protection. The equipment shall not be damaged by transient voltages as defined in MIL-STD-704 and shall automatically resume normal operation when the voltage returns within specified limits.

3.4 Performance.

3.4.1 General. Values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions.

3.4.2 Functions. The Test Set, Indicator AN/ARM-155 in conjunction with a test set, Radio, AN/ARM-156 shall be capable of performing the following functions in the maintenance of Radio Sets, AN/ARN-21, AN/ARN-52, AN/ARN-84, AN/ARN-86, AN/ARN-105, and other TACAN systems including C.N.I. (ASQ) units:

1. Indicate the Airborne TACAN Set range accuracy and search and track capabilities at different rates for each operating mode.

2. Indicate the Airborne TACAN Set bearing accuracy and search and track capabilities at different rates for each operating mode.

3. Monitor the identity tone.

4. Measure the Coupler, CU-395/ARN-21A range repeating accuracy.

5. Measure the CU-395 bearing differential and the fundamental resolver.

6. Measure the Azimuth Indicator, ID-307/ARN, bearing output accuracy on the dial, the bearing differential, the fundamental resolver, and the harmonic resolver.

7. Check output accuracy of displays and range potentiometers in CU-395 and ID-310.

8. Check operation of North Reference relay in CU-395.

9. Check output of phase detector circuits of CU-395 or CV-279, or ARN-52, ARN-84, ARN-86, ARN-105 or C.N.I. (ASQ) Units.

10. Indicate the proper presence and switching of the automatic antenna selection signal of the AN/ARN-84.

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11. Check the proper operation of the Air-to-air bearing capability of the AN/ARN-84.

12. Check the proper operation of the data link interface capability of the AN/ARN-84.

13. Check the operation of the AN/ARN-84 interfaces with AN/ASW-27 Data Link and the aircraft digital computer.

14. Duplicate the airborne TACAN Control Unit functions.

15. Enable fault isolation of malfunctions of the AN/ARN-84 of the correct Shop Replaceable Assembly (SRA).

16. In conjunction with the Synchro Range and Bearing Indicator (SRBI), permit measurement of bearing, distance, and heading indication accuracy and flag operation on any of the following instruments externally connected to the Test Set:

    ID-249, ID-250, ID-351, ID-357, ID-387, ID-388, AQU-2, AQU-4, ID-1013.

3.4.3 Tolerances

1. Range. The range indication provided by the Master Range Indicator shall be accurate within ±.05 nautical miles. The range portion of the Synchro Range and Bearing Indicator shall be accurate within ±.01 nautical miles.

2. Bearing. The bearing indications provided by the Master Bearing Indicator and the bearing portion of the Synchro Range and Bearing Indicator shall be accurate within ±.1 degree.

3. Range repeat. The range repeat indexing indication provided by the indicator panel shall be accurate within ±0.05 miles.

4. Bearing repeat. The bearing repeat indexing indication provided by the indicator panel shall be accurate within ±0.1 degree.

5. Deviation and to-from circuits. The measurement of Deviation and To-From meter indications shall be accurate within ±5 microamperes.

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3.5 Detail requirements.

3.5.1 Control indicator, C-8963/ARM-155. This Unit shall meet the following requirements:

3.5.1.1 Form factor. This unit when enclosed in its instrument case shall have maximum dimensions not to exceed 22-1/2 inches high by 20-1/2 inches wide by 20 inches long. The unit shall have a protective dust cover which will fit within the confines of the instrument case or a standard 19 inch relay rack. The unit when removed from the instrument case shall consist of a control panel 21 inches high by 19 inches wide and dust cover fitted behind the control panel not to exceed 19-5/8 inches high by 17-5/8 inches wide by 16-3/4 inches deep. The control panel shall be fitted with 18 captive screws for securing it to the instrument case, 12 of which will be used to secure the unit to a relay rack, as required by 3.3.4.1.

3.5.1.2 Weight. The weight of this unit shall not exceed 90 pounds.

3.5.1.3 Contents of unit. This unit shall contain the following circuits and functions:

3.5.1.3.1 Compass simulator. The Compass Simulator shall generate synchro stator voltages corresponding to simulated compass readings and provide the power necessary to operate the bearing differential synchro in the Indicator Coupler, CU-395/ARN-21A, Azimuth Indicator, ID-307/ARN, AN/ARN-52(V), AN/ARN-84, AN/ARN-86, AN/ARN-105, AN/ASQ-19, 56, 57, 58, and 88, in series combination with the relative bearing receiver synchro on the Bearing-Distance-Heading-Indicator ID-663 plus the parallel operation of the relative heading receiver synchro in the Course Indicator ID-387 or ID-351 and the compass and receiver synchro in the Course Indicator ID-250. A related dial shall be mounted to the front panel which shall provide indications from 0 through 359 degrees in increments of two degrees..

3.5.1.3.2 Test selector. A Test Selector switch shall be provided to permit selection of the various output signals of the CU-395, the CV-279, the ID-310 and the ID-307 as well as the DC Volts and the 380-420 Hz AC Volts to be measured by the meter. Those output signals to be measured as null indications at an index position shall include the bearing differential synchro output, the harmonic resolver synchro output, the fundamental resolver synchro output and the range transmitter synchro output. Those signals to be measured as amplitude measurements at various bearing settings shall include the output of the phase detector circuits in the CU-395 and the CV-279. The output of the range potentiometers in the ID-310 and CU-395 shall be measured as a null indication from wiper to wiper with the range circuit searching.

3.5.1.3.3 Synchro power. Synchro power of 26 volts and 12 volts, 400 Hz, shall be provided by means of a 115V to 26 and 12 volt transformer. The 12-volt power shall be 180° out of phase with the 26-volt power.

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3.5.1.3.4 Relays. Three relays shall be provided to perform the following functions related to control of the airborne TACAN set under test:

1. Apply Power

2. Enable transmit (T/R) mode of operation

3. Enable Air-to-air (A/A) mode of operation

3.5.1.3.5 North reference switching. A switch shall be provided to actuate the North Reference relay in the CU-395 North Reference position and to connect control signals to the ID-250 or ID-663 compass card and the bearing differential synchro input circuits of the airborne TACAN set or coupler or ID-307 under test, in the North Reference or Compass Simulator positions.

3.5.1.3.6 Bearing distance heading indicators. Three ID-663 indicators shall be provided. These shall be ID-663/U, or ID-663A/U but not ID-663B/U or ID-663C/U or other high impedance TACAN bearing input types. One shall be located on the front panel and the other two shall be located within the set to act as loads for testing.

3.5.1.3.7 Course indicator. An ID-387 shall be provided and located on the front panel.

3.5.1.3.8 Master bearing unit. An indicating unit shall be provided which is electrically similar to and interchangeable with the Bearing Indicator of the CU-395 Coupler or the ID-307 Azimuth Indicator. It shall not include the bearing repeating functions of the CU-395 and ID-307 or the harmonic resolver of the ID-307.

3.5.1.3.9 Master range unit. An indicating unit shall be provided which is electrically similar to and interchangeable with the Range Indicator in the CU-395 Coupler or the Range Indicator ID-310. It shall not include the flag control circuits and take off potentiometers of the CU-395 and ID-310 or the range repeating functions of the CU-395.

3.5.1.3.10 Composite control unit. A Composite Control Unit shall provide circuitry for control of the AN/ARN-21, AN/ARN-52, AN/ARN-84 and AN/ARN-105, and shall plug into a suitable receptacle recessed in the front panel. Adapters shall be supplied to permit substituting control units from AN/ARN-21 or AN/ARN-52 or AN/ARN-84 TACAN.

3.5.1.3.10.1 Interlock provision. There shall be a device to disable electrical power to the Test Set, Indicator if a wrong combination of external control unit and control unit adapter is selected.

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3.5.1.3.11 Synchro range and bearing indicator (SRBI). A digital indicator shall be provided to display range and bearing separately and concurrently. Range shall be displayed in hundreds, tens, units, tenths and hundredths of nautical miles to an accuracy of ±0.01 nautical miles. Bearing shall be displayed in hundreds, tens, units, and tenths of degrees to an accuracy of ±0.1 degree. These displays shall be computed from the Range and Bearing Synchro output signals provided by the Airborne TACAN Set under test for driving synchro repeater indicators. The load impedances presented by the input circuits of the SRBI shall each be at least 100,000 ohms impedance to ground. This indicator shall be used concurrently with and in addition to the one, two or three instrument loads selected by the Load Selector switch of 3.5.1.4.2. The internal harness of the Control Indicator shall provide the capability of driving indicators, ID-250 and ID-388, when such instruments are mounted in the front panel as replacement for ID-663 and SRBI respectively.

3.5.1.3.12 Meter circuit. A meter circuit shall be provided consisting of a 0-200 microammeter, a series diode, series and shunt resistors, a sensitivity switch and a polarity switch. A DC null shall be obtainable for the various repeat and resolver positions of the Test Selector. A load, equivalent to the Deviation and To-From meters of three ID-387 or ID-351 Course Indicators shall be obtained for all positions of the Test Selector including the Deviation and To-From positions, and the meter shall check tracking accuracy of the range take-off potentiometers in the ID-310 or CU-395 in Range Pot. position of the selector. Circuits will be provided to enable measurement of the following utilizing this microammeter:

DC Volts	Deviation Full Scale accuracy
380-420 Hz AC Volts	To-From Zero accuracy
Bearing Repeat accuracy	To-From Full Scale accuracy
Fundamental Resolver accuracy	Range Repeat accuracy
Harmonic Resolver accuracy	Range Pot accuracy
Deviation Zero accuracy	Off

3.5.1.3.13 Time totalizing meter. A meter shall be provided to indicate total elapsed operating hours as specified by MIL-M-7793.

3.5.1.3.14 Standby heater. The standby heater, energized when the 380-420 Hz Power Switch is in the OFF position, shall be capable of raising the average temperature of the unit to at least 6° C above ambient with the unit outside its transit case. A front panel white light shall be provided to indicate that the standby heater is energized.

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3.5.1.4 Controls. The following controls shall be located on the front panel of this unit.

3.5.1.4.1 Test selector switch. A twelve-position rotary test selector switch shall be provided for selecting the type of test to be conducted.

3.5.1.4.2 Load selector switch. Three-position switch to select one, two, or three indicators as loads for the Range and Bearing Synchro output circuits of the Airborne TACAN Set under test.

3.5.1.4.3 North reference - compass simulator switch. Two-position toggle switch for controlling the functions described in 3.5.1.3.5.

3.5.1.4.4 Meter sensitivity switch. Momentary—ON type two-position toggle switch for increasing meter sensitivity.

3.5.1.4.5 Meter polarity switch. Two-position toggle switch for reversing direction of current flow through the meter.

3.5.1.4.6 Compass simulator. A continuously variable control to mechanically position the rotor of a synchro transmitter to permit selection of the desired bearing.

3.5.1.4.7 380-420 Hz power switch. Two-position toggle switch for applying power to the Test Set and Airborne TACAN Set under test in the ON position. In the OFF position, power shall be applied to the Standby Heater.

3.5.1.4.8 28 VDC power switch. Two-position toggle switch for applying 28 VDC to the Test Set and the Airborne TACAN Set under test.

3.5.1.4.9 Aircraft-master switch. An Aircraft-Master Switch shall be provided to select an aircraft system of indicators including a Coupler CU-395, an ID-307 Azimuth Indicator, an ID-310 Range Indicator or to select a system or master indicators including a Master Bearing Indicator and a Master Range Indicator.

3.5.1.4.10 Heater power overload reset switch. A circuit breaker shall be provided to protect the heater power circuit in the event of an overload. A push to reset feature shall be provided.

3.5.1.4.11 Panel lighting switch. A two-position toggle switch for applying 28 VDC or 5 VDC for lighting AN/ARN-84 control box.

3.5.1.5 Front panel indicators. The following indicators shall be located on the front panel:

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1. 380-420 Hz Power (Red Light).

2. 28 VDC Power (Red Light).

3. Heater Power (White Light).

4. Antenna Switching Signal Indicator (Electro-mechanical).

3.5.1.6 Electrical connections. The connectors provided on the front panel shall conform to MIL-T-21200, except as noted by an asterisk (*) in Table I.

3.5.2 Transit case, control-indicator, Test Set, AN/ARM-155. The transit case shall meet the following requirements:

3.5.2.1 Function. This case shall provide compartments for the storage of the equipment, handbooks and accessories.

3.5.2.2 Form factor. The dimensions of the test transit case shall not exceed 28 inches wide, 20 inches high and 32 inches long.

3.5.2.3 Weight. The weight of the transit case shall not exceed 61 pounds.

3.5.2.4 Construction. The transit case shall be designed and constructed to meet the applicable requirements of MIL-T-21200 for transit cases.

3.5.3 Indicator control, ID-1812/ARM-155. This unit shall contain power supplies, indicators and controls and shall provide test signals in addition to those signals provided by Test Set, Radio, AN/ARM-156 required to monitor the performance, and to fault isolate failures to the SRA level, of the AN/ARN-84 TACAN. This unit shall meet the following requirements.

3.5.3.1 Form factor. This unit when enclosed in its instrument case shall have maximum dimensions not to exceed 19 inches high by 20-1/2 inches wide by 20 inches long. The unit shall have a protective dust cover which will fit within the confines of the instrument case or a standard 19 inch relay rack. The unit, when removed from the instrument case, shall consist of a control panel not to exceed 17-1/2 inches high by 19 inches wide and dust cover fitted behind the control panel not to exceed 16-1/8 inches high by 17-5/8 inches wide by 16-3/4 inches deep. The control panel shall be fitted with captive screws for securing it to the instrument case.

3.5.3.2 Weight. The weight of the equipment shall not exceed 65 pounds.

3.5.3.3 Contents of unit. This unit shall contain the following circuits:

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3.5.3.3.1 Low voltage power supplies. The power supplies furnished shall provide supply voltages necessary to operate the circuitry of this unit.

3.5.3.3.2 AN/ASW-27 data link displays. Circuitry shall be provided to convert AN/ARN-84 output analog range and bearing and binary station number from the AN/ASW-27 Data Link Interface to a digital format. Decimal displays shall be provided to display range from 000.00 to 299.99 nautical miles to an accuracy of ±0.01 nautical miles, bearing from 000.00 to 359.9 degrees of an accuracy of ±0.1 degree, and station number from 1 to 126. All power for the Data Link Displays is provided by the power supplies of 3.5.3.3.1.

3.5.3.3.3 Digital output indicator section. Circuitry shall be provided to generate five interrogation signals as specified in MIL-N-81207A(AS), which shall be routed to the Airborne TACAN Set under test as shown below:

ITEM	SIGNAL	LINE	MNEMONIC
1	Read	Read Address Line	RAL
2	Address A	Address Line A	ALA
3	Address B	Address Line B	ALB
4	Shift clock	Data Shift Clock	DSC
5	Shift Clock Complement	Data Shift Clock Complement	DSC

3.5.3.3.3.1 Decimal conversion.

1. Circuitry shall be provided to convert the serial data range, bearing, and channel number words from the AN/ARN-84 Converter, to decimal format.

2. Circuitry shall be provided to convert the parallel data range and bearing words from the AN/ARN-84 Receiver-Transmitter, to a decimal format..

3.5.3.3.4 Continuous monitors. Circuitry shall be provided to allow display of NO-GO indicators for the TACAN Receiver-Transmitter unit, the Control Unit, the Converter Unit and the Antenna.

3.5.3.3.5 Status indicators section. Circuitry shall be provided to generate Bearing Antenna Status, both GO and NO-GO. Circuitry shall also be provided to generate a TACAN Self-Test Command signal.

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3.5.3.3.6 TACAN suppression. Circuitry shall be provided to generate the external blanking signal specified in MIL-N-81207(AS).

3.5.3.3.7 VAST signature resistors. Means of selection of the VAST signature resistors of the AN/ARN-84 for measurement at the Test Signal Output Jacks, J201 and J202, shall be provided.

3.5.3.4 Controls. The following controls shall be located on the front panel.

3.5.3.4.1 TACAN suppression. A three-position toggle switch shall be provided to permit selection of pulsed, momentary (MON), or no suppression (OFF) of AN/ARN-84 transmission.

3.5.3.4.2 Voltage selector. A ten-position rotary switch shall be provided to permit monitoring of AN/ARN-84 Internal voltages.

3.5.3.4.3 Data link. Controls shall be provided to (1) enable data link transmission either automatically or manually and (2) select data link pulses received by the airborne TACAN set or by the SIMULATOR for display.

3.5.3.4.4 Self-test. A three-position switch shall be provided to permit selection of programmed self-test of Displays, Standard Internal, (STD INT), or OFF.

3.5.3.4.5 Signal selection. Two output controls shall be provided to allow selection of any of 24 test signals as outputs at either of two front panel connectors.

3.5.3.4.6 Sync selector. A control shall be provided to allow selection of any of four sync output signals from the test set.

3.5.3.4.7 Test selector. A control shall be provided to select any of six AN/ARN-84 test configurations and internal self test.

3.5.3.4.8 TACAN BIT (Self-Test). A three-position switch shall be provided to permit the selection of the AN/ARN-84 BIT (Self-Test) ON, OFF or momentary ON (MOM).

3.5.3.4.9 Antenna status. A switch shall be provided to simulate an antenna NO-GO status.

3.5.3.4.10 Data selector. A switch shall be provided to permit the selection of the display of the RANGE, BEARING and CHANNEL information from either the Receiver-Transmitter, Converter or ASW-27 Interface.

3.5.3.5 Front panel indicators and displays. The following displays and indicators shall be located on the front panel:

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3.5.3.5.1 Digital output indicators. The inputs to these displays shall be internally computed from the digital word outputs provided by the AN/ASW-27 Data Link Interface the Digital Computer Interface of the Converter and the GSE Connectors of the Receiver-Transmitter of the AN/ARN-84 Airborne TACAN set under test:

1. Range. Range shall be displayed in decimal form as follows:

   AN/ASW-27 Interface - 0.00 to 299.99 NM
   Digital Computer Interface - 0.000 to 299.975 NM
   R/T GSE Connectors - 0.0000 to 299.9875 NM

2. Bearing. Bearing shall be displayed in decimal form as follows:

   AN/ASW-27 Interface - 000.0 to 259.9 deg.
   Digital Computer Interface - 000.00 to 359.75 deg.
   R/T GSE Connectors - 000.000 to 359.875 deg.

3. Channel. Channel number shall be displayed in decimal form from 1 to 126.

4. Valid. Valid indicators shall be provided for RANGE, BEARING and CHANNEL.

3.5.3.5.2 NO-GO indicators. NO-GO displays shall be provided to continuously monitor the Receiver-Transmitter, Control Unit, Converter and the Antenna.

3.5.3.5.3 Status indicators. Indicators shall be provided to indicate GO status for:

1. Bearing Antenna
2. Frequency Synthesizer
3. Control Unit
4. Power Supply
5. Range Lock
6. Bearing Lock
7. Frequency Synthesizer Lock
8. Range Coupler
9. Bearing Coupler
10. BIT Status

3.5.3.5.4 TACAN data link displays. Decimal indicators shall be provided to display:

1. Transmitted. Four digit count of data link pulses transmitted.

2. Received. Four digit count of data link pulses received.

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3.5.3.5.5 Power supply voltmeter. A voltmeter shall display switch selected voltages from within the airborne TACAN set under test or within the indicator Control during Internal Test. The switch selected voltages shall appear at the voltage Monitor Output connector (J204) provided on the front panel.

3.5.3.5.6 Time totalizing meter. A meter shall be provided to indicate total elapsed operating hours as specified by MIL-M-7793.

3.5.3.5.7 TACAN voltage status indicator. A Status Indicator shall be provided which shall indicate VALID when the following DC voltages are within acceptable limits: +5V, +5.2V, -5.2V, +8.5V, +12V, -12V, +28V, +30V, +38V and high voltages.

3.5.3.6 Electrical connections. The external connectors provided are listed in Table II and shall conform to MIL-T-21200.

TABLE II

LIST OF CONNECTORS OF INDICATOR UNIT

FRONT PANEL MOUNTED CONNECTORS NAME	TYPE	CONNECTOR DESIGNATION
Sync Out	BNC	J203
Output Number 1	BNC	J201
Output Number 2	BNC	J202
Data Link Enable Input	BNC	J206
Data Link Input	BNC	J208
Interrogation Monitor Input	BNC	J205
TACAN Video Output	BNC	J207
Voltage Monitor Output	BNC	J204
TACAN Suppression (SPI) Output	BNC	J209
TACAN Suppression (SPO) Input	BNC	J210
North Reference Trigger Input	BNC	J211

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TABLE II (Continued)

FRONT PANEL MOUNTED CONNECTORS NAME	TYPE	CONNECTOR DESIGNATION
Quantized Analog Bearing Pulse Output	BNC	J212
Quantized Analog Range Pulse Output	BNC	J214
TACAN Bearing Clock Output	BNC	J215
Quantized Bearing Reference Pulse Output	BNC	J213
AN/ARN-84 Interface	Multi-Pin	J218
From AN/ARN-84, J4	Multi-Pin	J216
From AN/ARN-84, J5	Multi-Pin	J217
C-8963/ARM-155 Interconnect	Multi-Pin	J219
REAR PANEL MOUNTED CONNECTORS NAME	TYPE	
0.005 Nautical Mile Clock Output/Input	BNC	

3.5.3.7 Internal-external clock selection. A rear-panel mounted two position switch shall be provided to enable selection of either of two 0.005 nautical mile clock sources.

3.5.4 Transit Case, Indicator, Control, Test Set, AN/ARM-155. The transit case shall meet the following requirements:

3.5.4.1 Function. This case shall provide compartments for the storage of the equipment, handbooks, and accessories.

3.5.4.2 Form factor. The dimensions of the transit case shall not exceed 28 inches wide, 27 inches high and 32 inches long.

3.5.4.3 Weight. The weight of the transit case shall not exceed 60 pounds.

3.5.4.4 Construction. The transit case shall be designed and constructed to meet the applicable requirements of MIL-T-21200 for transit cases.

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3.5.5 Mounting base, MT-4410/ARM-155. The fixture shall be designed to securely hold the AN/ARN-84 TACAN while the unit is undergoing testing. The Mounting Base will aid in functional testing and fault isolation of the airborne TACAN set and shall meet the following requirements.

3.5.5.1 Form factor. The mounting base shall have maximum dimensions not to exceed 12 inches high by 10 inches wide by 24 inches long.

3.5.5.2 Weight. The weight of the mounting base shall not exceed 9 pounds.

3.5.5.3 Contents. The mounting base, MT-4410/ARM-155 shall contain the following features as found in Base, Shock Mount, Electrical Equipment, MT-4354/ARN-84(V):

1. Line Filter, FL-1.

2. Diode, CR2.

3. Connectors, J1901, J1902, J1905, J1906, J1907, and X2P1.

3.5.6 Transit case, mounting base, electrical equipment, AN/ARM-155. The transit case shall meet the following requirements:

3.5.6.1 Function. This case shall provide compartments for the storage of the equipment and accessories.

3.5.6.2 Form factor. The dimensions of the transit case shall not exceed 28 inches wide, 27 inches high and 32 inches long.

3.5.6.3 Weight. The weight of the transit case shall not exceed 77 pounds.

3.5.6.4 Construction. The transit case shall be designed and constructed to meet the applicable requirements of MIL-T-21200 for transit cases.

3.5.7 Cable assemblies and adapters.

3.5.7.1 Cable and adapters supplied. The following are cable assemblies which shall be supplied for use with the Test Set, Indicator, AN/ARM-155 for testing the RT-1022/ARN-84(V).

3.5.7.1.1 Cable assembly, power, electrical-115 VAC W-7 (8'0"). An A-C power cord shall be supplied consisting of a 96-inch length of three-conductor #18 AWG flexible power cable. The cable shall be terminated at one end by a Type MS3106A10SL3S connector and at the other end by a Type MS3107A20-4P connector.

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This cable shall connect 115 Volts A.C., 380-420 Hertz primary power to the C-8963/ARM-155.

3.5.7.1.2 Cable assembly, power, electrical - +28 VDC W-9 (8'0"). This D-C power cord shall be supplied consisting of a 96-inch length of two-conductor #16 AWG flexible power cable. It shall be terminated on one end by a Type MS3106A12S-3S connector and at the other end by a Type MS3107A16-11P connector. This cable shall supply +28 Volts D.C. primary power to the C-8963/ARM-155.

3.5.7.1.3 Cable assembly, special purpose, electrical W-12 (4'5"). An interconnect cable shall be supplied consisting of a 53-inch length of 23-conductor, flexible, jacketed, shielded cable, with concentrically wound layers of insulated wire. Twenty-one of the conductors will be 22 AWG and the remaining two will be 16 AWG. The cable shall be terminated on one end by a Type MS27473T16B99P connector and at the other end by a Type MS27473T16B99S connector. This is the long interconnect cable which is used to interconnect the C-8963/ARM-155 and the ID-1812/ARM-155 units when they are separated. The cable provides 4-1/2 feet separation between the units.

3.5.7.1.4 Cable assembly, special purpose, electrical W-13 (8'10"). This cable shall be supplied consisting of a 106-inch length, 55-conductor #22 AWG shielded cable. This cable shall be terminated on one end by a Type MS27473T16B55S connector and on the other end by a Type MS27473T16B55P connector. This cable shall connect the AN/ARN-84(V) TACAN digital range and bearing signals from the MT-4410/ARM-155 to the ID-1812/ARM-155.

3.5.7.1.5 Cable assembly, special purpose electrical branched W-14 (8'10"). This cable shall be supplied consisting of a 106-inch length, branched, 17-conductor, shielded cable. Three conductors will be 18 AWG, seven conductors 20 AWG, and seven conductors shall be 24 AWG. The cable shall be terminated on one end by a Type MS3106A18-9S (or a MS3106R18-9S) connector, on one end by a Type MS3106R18-1P connector and at the other end by a Type 8818600004-7 connector. This cable shall connect power and control signals between the C-8963/ARM-155 and the MT-4410/ARM-155.

3.5.7.1.6 Cable assembly, special purpose electrical W-15 (8'10"). This cable shall be supplied consisting of a 106-inch length, 85-conductor, flexible, jacketed, shielded cable, with concentrically wound layers of 24 AWG insulated wire. The cable will be terminated by a M81511/06E18-( ) connector at each end. This cable shall connect connector J4 of the RT-1022/ARN-84(V) with connector J216 of ID-1812/ARM-155.

3.5.7.1.7 Cable assembly, special purpose electrical W-16 (8'10"). This cable shall be supplied consisting of a 106-inch length, 85-conductor, flexible, jacketed, shielded cable, with concentrically wound layers of 24 AWG insulated wire.

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The cable will be terminated by a M81511/06E18-( ) connector at each end. This cable shall connect connector J5 of the RT-1022/ARN-84 (V) with connector J217 of ID-1812/ARM-155.

3.5.7.1.8 Cable assembly, radio frequency W-17 (9'0"). This cable shall be supplied consisting of a 108-inch length of RG-58/U R.F. Cable terminated on each end by a M39012/16-0004 connector. This cable shall connect the Suppression Pulse Output (SPO) connector J1026 of the RT-1022/ARN-84(V) with connector J210 of ID-1812/ARM-155.

3.5.7.1.9 Cable assembly, radio frequency W-18 (9'0"). This cable shall be supplied consisting of a 108-inch length of RG-58/U R.F. Cable terminated on each end by a M39012/16-0004 connector. This cable shall connect the Suppression Pulse Input (SPI) connector J1027 of the RT-1022/ARN-84(V) with connector J209 of ID-1812/ARM-155.

3.5.7.1.10 Cable assembly, radio frequency W-19 (9'0"). This cable shall be supplied consisting of a 108-inch length of RG-58/U R.F. Cable terminated on each end by a M39012/16-0004 connector. This cable shall connect the North Reference Trigger (NRT) connector J18 of the PP-6730/ARM-156 with connector J211 of ID-1812/ARM-155.

3.5.7.1.11 Cable assembly, radio frequency W-20 (9'0"). This cable shall be supplied consisting of a 108-inch length of RG-58U R.F. Cable terminated on each end by a M39012/16-0004 connector. This cable shall connect the External Transmit Enable (ETE) connector J12 of the PP-6730/ARM-156 with connector J206 of ID-1812/ARM-155.

3.5.7.1.12 Cable assembly, radio frequency W-21 (9'0"). This cable shall be supplied consisting of a 108-inch length of RG-58/U R.F. Cable terminated on each end by a M39012/16-0004 connector. This cable shall connect the External Transmit Pulse (ETP) connector J13 of the PP-6730/ARM-156 with connector J208 of ID-1812/ARM-155.

3.5.7.1.13 Cable assembly, radio frequency W-22 (9'0"). This cable shall be supplied consisting of a 108-inch length of RG-58/U R.F. Cable terminated on each end by a M39012/16-004 connector. This cable shall connect the Data Link Interrogate Monitor signal (DLIM) connector J4 of the SG-980/ARM-156 with connector J205 of ID-1812/ARM-155.

3.5.7.1.14 Cable assembly, special purpose, electrical, branched W-23 (11'1.5"). This cable assembly shall be supplied consisting of an ID-663 Test Box which has two switches, a voltage regulator and three cables wired to it. A 108-inch length of nineteen #22 AWG conductors and seven #24 AWG conductors is terminated on one end by connector P2 with Type 881860004-1 connector, and is terminated on the other end by the ID-663 Test Box. Connector P2 of W-23/ARM-155

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connects to connector J903 of C-8963/ARM-155. A 108-inch length of nineteen #22 AWG conductors and seven #24 AWG conductors is terminated on one end by connector P3 with a Type 8818600004-2 connector, and is terminated on the other end by the ID-663 Test Box. Connector P3 of W-23/ARM-155 connects to J907 of C-8963/ARM-155. An 18-inch length of thirty #22 AWG conductors is terminated on one end by connector P1 with a Type MS24236F21-30N connector, and is terminated on the other end by the ID-663 Test Box. Connector P1 of W-23/ARM-155 connects to an ID-663(*)/U Bearing-Distance-Heading Indicator (BDHI).

The W-23/ARM-155 Cable provides the capability of testing those functions of the Bearing-Distance-Heading Indicator that are normally exercised by a TACAN system.

3.5.7.1.15 Cable assembly, special purpose, electrical branched W-24 (11'1.5"). This cable assembly shall be supplied consisting of an ID-1013 Test Box which has 6 switches, a resistor, 26 test jacks, and 5 cables wired to it. A 108-inch length of fifteen #22 AWG conductors is terminated on one end by connector P3 with a Type 8818600004-8 connector (a 32-pin connector) and is terminated on the other end by the ID-1013 Test Box. Eleven additional #22 AWG conductors are terminated inside the P3 connector but are not electrically connected to it. The eleven additional conductors are terminated at the other end inside the ID-1013 Test Box but are not electrically connected to it. Connector P3 of W-24/ARM-155 connects to J901 of C-8963/ARM-155. A 108-inch length of fourteen #22AWG conductors is terminated on one end by connector P4 with a Type 881860004-1 connector (a 24-pin connector) and is terminated on the other end by the ID-1013 Test Box. Twelve additional #22 AWG conductors are terminated inside the P4 connector but are not electrically connected to it. The twelve additional conductors are terminated at their other end inside the ID-1013 Test Box but are not electrically connected to it. Connector P4 of W-24/ARM-155 connects to J903 of C-8953/ARM-155. A 108-inch length of thirteen #22 AWG conductors is terminated on one end by connector P5 with a Type 8818600004-2 connector (a 24-pin connector) and is terminated on the other end by the ID-1013 Test Box. Thirteen additional #22 AWG conductors are terminated inside the P5 connector but are not electrically connected to it. The thirteen additional conductors are terminated at their other end inside the ID-1013 Test Box but are not electrically connected to it. Connector P5 of W-24/ARM-155 connects to J907 of C-8963/ARM-155. An 18-inch length of thirty-one #22 AWG conductors is terminated on one end by connector P1 with a Type MS3116F18-32S connector (a 32-pin connector) and is terminated on the other end by the ID-1013 Test Box. Connector P1 of W-24/ARM-155 connects to J2 of either ID-1013 (*) A or AF/A24J-1. An 18-inch length of fifty-five #22 AWG conductors is terminated on one end by connector P2 with a Type MS3116F22-55S connector (a 55-pin connector) and is terminated on the other end by the ID-1013 Test Box. Connector P2 of W-24/ARM-155 connects to J1 of either ID-1013 (*) A or AF/A24J-1.

The W-24/ARM-155 Cable provides the capability of testing those functions of the ID-1013(*) A and AF/A24J-1 Horizontal Situation Indicators that are normally exercised by a TACAN system.

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3.5.7.1.16 Cable assembly, special purpose, electrical branched W-25 (11'1.5"). This cable assembly shall be supplied consisting of an AQU Test Box which has 3 switches, a resistor, a transformer, 12 test jacks and 4 cables wired to it. A 108-inch length of fourteen #22 AWG conductors (7 are shielded) is terminated on one end by connector P2 with a Type 8818600004-8 (a 32-pin connector) and is terminated on the other end by the AQU Test Box. Twelve additional #22 AWG conductors are terminated inside connector P2 but are not electrically connected to it. The twelve additional conductors are terminated at the other end by the AQU Test Box but are not electrically connected to it. Connector P2 of W-25/ARM-155 connects to J901 of C-8963/ARM-155. A 108-inch length of thirteen #22 AWG conductors (6 are shielded) is terminated on one end by connector P3 with a Type 8818600004-1 (a 24-pin connector) and is terminated on the other end by the AQU Test Box. Thirteen additional #22 AWG conductors are terminated inside connector P3 but are not electrically connected to it. The thirteen additional conductors are terminated at the other end by the AQU Test Box but are not electronically connected to it. Connector P3 of W-25/ARM-155 connects to J903 of C-8963/ARM-155. A 108-inch length of nine #22 AWG conductors and four #24 AWG shielded conductors is terminated on one end by connector P4 with a Type 8818600004-2 (a 24-pin connector) and is terminated on the other end by the AQU Test Box. Thirteen additional conductors are terminated inside connector P4 on one end and the AQU Test Box at their other end, but are not electrically connected to the connector or test box. Connector P4 of W-25/ARM-155 connects to J907 of C-8963/ARM-155. An 18-inch length of fifty-five #22 AWG conductors (13 are shielded) is terminated on one end by connector P1 with a Type MS3116F22-55S (a 55-pin connector) and is terminated on the other end by the AQU Test Box. Connector P1 of W-25/ARM-155 connects to J1 of the AQU-2/A, AQU-4/A, or AQU-6/A Horizontal Situation Indicator. The W-25 Cable Assembly provides the capability of testing those functions of the Horizontal Situation Indicator that are normally exercised by a TACAN system.

3.5.7.1.17 Cable assembly, power, electrical, branched W-26 (8'10"). This cable shall be 106 inches in length consisting of 43 conductors which are terminated by 2 connectors on one end and 3 connectors on the other. Sixteen conductors are terminated in connector P1 by a Type 8818600004-9 (32-pin connector). Twelve of the conductors are #22 AWG (4 are shielded), one is #20 AWG, and three are #16 AWG (1 is shielded). Connector P1 of W-26/ARM-155 connects to J908 of C-8963/ARM-155.

Twelve conductors are terminated in connector P2 by a Type 8818600004-3 (24-pin connector). Eleven of the conductors are #22 AWG (5 are shielded) and one conductor is #16 AWG. Connector P2 of W-26/ARM-155 connects to J910 of C-8963/ARM-155.

Twelve conductors are terminated in connector P3 by a Type 8818600004-6 (24-pin connector). Nine of the conductors are #22 AWG, two are #20 AWG, and one is #16 AWG. Connector P3 of W-26/ARM-155 connects to J911 of C-8963/ARM-155.

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Eighteen conductors are terminated in connector P4 by a Type MS3106R22-14S (20-pin connector). Thirteen of the conductors are #22 AWG (2 are shielded), two are #20 AWG, and three are #16 AWG. Connector P4 of W-26/ARM-155 connects to J1902 of MT-4410/ARM-155.

Twenty-three conductors are terminated in connector P5 by a Type MS3106R24-28P (24-pin connector). Nineteen of the conductors are #22 AWG (7 are shielded), two are #20 AWG, and two are #16 AWG (1 is shielded). Connector P5 of W-26/AN/ARM-155 connects to J1905 of MT-4410/ARM-155.

Additional conductors are terminated but not electrically connected to the connectors as follows: 2 in P1, 1 in P2, 1 in P4, and 1 in connector P5.

The W-26 Cable Assembly connects the AN/ARN-84(V) TACAN analog range and bearing signals to the Indicator Test Set Analog Unit C-8963/ARM-155.

3.5.7.1.18 Cable assembly, special purpose, electrical W-27 (1'6"). An interconnect cable shall be supplied consisting of an 18-inch length of 23 conductor, flexible, jacketed, shielded cable, with concentrically wound layers of insulated wire. Twenty-one of the conductors will be #22 AWG and the remaining two will be #16 AWG. The cable shall be terminated on one end by a Type MS27473T16B99P connector and at the other end by a Type MS27473T16B99S connector. This si the short interconnect cable which is used to interconnect the C-8963/ARM-155 and the ID-1812/ARM-155 when they are stacked together.

3.5.7.1.19 Adapter, connector (course adapter). This adapter shall have a Type 8818600010-2 (a 32-pin connector) with the inputs and outputs shorted together, The adapter shall connect to J901 of C-8963/ARM-155. This will permit the use of the C-8963/ARM-155's ID-387 Course Indicator when testing a TACAN.

3.5.7.1.20 Adapter, connector (bearing adapter). This adapter shall have a Type 8818600010-1 (24-pin connector) with the inputs and outputs shorted together. The adapter shall connect to J903 of C-8963/ARM-155 which will connect the ID-663 of C-8963/ARM-155 to the bearing output of the TACAN under test.

3.5.7.1.21 Adapter, connector (range adapter). This adapter shall have a Type 8818600010-3 (24-pin connector) with the inputs and outputs shorted together. The adapter shall connect to J907 of C-8963/ARM-155 which will connect the ID-663 of C-8963/ARM-155 to the range output of the TACAN under test.

3.5.7.1.22 Adapter, connector (control unit ARN/84). This adapter shall have a Type MS3102R20-27S connector P25 and a Type 8818600009-1 connector P1. P1 will connect with the AN/ARN-84(V) Control Unit. P25 will connect with J925 of C-8963/ARM-155. The purpose of the adapter is to permit testing of the AN/ARN-84(V) Control Unit.

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3.5.7.1.23 Adapter, connector (control unit ARN-21/52). This adapter shall have a Type MS3102R20-27S connector P22 and a Type 8818600009-1 connector P1. P1 will connect with the AN/ARN-52(V) or AN/ARN-21 Control Unit. P22 will connect with J922 of C-8963/ARM-155. The purpose of the adapter is to permit testing of the C-2010/ARN-52(V), C-3928/ARN-52(V), C-7893/ARN-52(V), C-7712/ARN-52(V), C-1763/ARN-21A or C-886/ARN-21 Control Units.

3.5.7.2 Cables and test bench harnesses not supplied. The following cables and test bench harnesses, in addition to the items of 3.5.7.1, are used with the Test Set Indicator AN/ARM-155. These cables and test bench harnesses are used for testing airborne TACAN sets and indicators other than the AN/ARN-84(V) airborne TACAN set, ID-663, AQU-4, and ID-1013 instruments.

3.5.7.2.1 Special purpose electrical cable assembly W-11/ARM-31 (8'6"). This cable shall be 102 inches in length, consisting of 60 wires with suitable outer protective coating as follows: 46 wires of No. 22 gauge and 14 shielded wires of No. 22 gauge. This cable shall be terminated at one end by an assembly of two ribbon-type 24-pin connectors (Amphenol 26-4303 - 24P and 26-4306 - 24P or approved equivalents) and two ribbon-type 32-pin connectors (Amphenol 26-4302 - 32P and 26-4303 - 32P or approved equivalents) and at the other end by a Type AN 3106B-36-8W. One end of this cable will connect to J908, J909, J910 and J911 of C-8963/ARM-155. The other end of the cable will connect to J201 and J202 of CU-395/ARN-21. The purpose of the cable assembly is to permit testing of the CU-395/ARN-21A Indicator Coupler.

3.5.7.2.2 Special purpose electrical cable assembly W-1/ARM-31(4'0"). This cable shall be 48 inches in length consisting of 7 wires with suitable outer protective coating as follows: 5 wires of No. 22 gauge and 2 shielded wires of No. 22 gauge. This cable shall be terminated at one end with a ribbon-type connector (Amphenol 26-4303 - 24P or an approved equivalent) and at the other end with a Type AN3106-18 -1S connector. One end of the cable will connect to J910 of C-8963/ARM-155 and the other end to J1501 of CV-279/ARN-21A. The purpose of the cable is to permit testing of the CV-279/ARN-21A Phase Detecting Network.

3.5.7.2.3 Special purpose electrical cable assembly W-2/ARM-31 (8'6"). This cable shall be 102 inches in length consisting of 42 wires with suitable outer protective coating as follows: 26 wires of No. 22 gauge, 14 of shielded No. 22 gauge and 2 wires of No. 18 gauge. This cable shall be terminated at one end by an assembly of two ribbon-type connectors (Amphenol 26-4504 - 24P and 26-4505 - 24P or approved equivalents) and at the other end by a 45-terminal female connector (Cannon DPD-45-33S-1 or an approved equivalent). One end of the cable will connect to J912 and J913 of C-8963/ARM-155. The other end will connect to the RT-220/ARN-21 Radio Receiver-Transmitter.

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3.5.7.2.4 Special purpose electrical cable assembly W-3/ARM-31 (4'0"). This cable shall be 48 inches in length consisting of 30 wires with suitable outer protective coating as follows: 18 wires of No. 22 gauge and 12 shielded wires of No. 22 gauge. This cable shall be terminated at one end with a ribbon-type connector (Amphenol 26-4302-32P or an approved equivalent) and at the other end with a Type AN3106A-28-12S connector. One end of the cable will connect to J908 of C-8963/ARM-155 and the other end to J101 of ID-307/ARN. The purpose of the cable is to permit testing of the ID-307/ARN Azimuth Indicator.

3.5.7.2.5 Special purpose electrical cable assembly W-4/ARM-31 (4'0"). This cable shall be 48 inches in length, consisting of 18 wires of No. 22 gauge with suitable outer protective coating. This cable may be terminated at one end by a ribbon-type 32-terminal connector (Amphenol 26-43-1-32P or an approved equivalent) and at the other by a Type AN3106A-28-21S connector. One end of the cable will connect to J901 of C-8963/ARM-155 and the other end to the Course Indicator. The purpose of the cable is to permit testing of the ID-249 (*)/ARN, ID-351 (*)/ARN, ID-357/ARN, and ID-387/ARN Course Indicators.

3.5.7.2.6 Special purpose electrical cable assembly W-5/ARM-31 (4'0"). This cable shall be 48 inches in length, consisting of 9 wires of No. 22 gauge with suitable outer protective coating. This cable may be terminated at the one end by a ribbon-type 24-terminal connector (Amphenol 26-4301-24P or an approved equivalent) and at the other end by a Type AN3106A-22-14S connector. One end of the cable will connect to J903 of C-8963/ARM-155 and the other end to a ID-250 (*)/ARN Course Indicator.

3.5.7.2.7 Special purpose electrical cable assembly W-6/ARM-31(4'0"). This cable shall be 48 inches in length, consisting of 10 wires of No. 22 gauge with suitable outer protective coating. This cable may be terminated at one end by a ribbon-type 24-terminal connector (Amphenol 26-430Z-24P or an approved equivalent) and at the other by a Type AN3106A-20-27S connector. One end of this cable will connect to J907 of C-8963/ARM-155 and the other end to a ID-388/APN Range Indicator.

3.5.7.2.8 Special purpose electrical cable assembly W-10/ARM-31 (4'0"). This cable shall be 48 inches in length consisting of 26 wires with suitable outer protective coating as follows: 18 wires of No. 22 gauge and 8 shielded wires of No. 22 gauge. This cable shall be terminated at one end with a ribbon-type connector (Amphenol AN26-4303-32P or an approved equivalent) and at the other end with a Type AN3106A-28-12S connector. One end of this cable will connect to J909 of C-8963/ARM-155 and the other end to J1601 of a ID-310 (*)/ARN Range Indicator.

3.5.7.2.9 Test cable, Republic 54502040, RT1030/ARN-105-ARM-31 (6'0"). This cable shall be 72 inches in length consisting of two ITT Cannon connectors on one end, a 7120 MS3116F20-41P (41-pin) and a 7120 MS3116F14-5S (5-pin) which connects to the AN/ARN-105 TACAN. The five Amphenol connectors on the other end mate with C-8963/ARM-155 as follows: 26-4304-24P (24-pin) mates with J912;

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26-4303-24P (24-pin) mates with J910; 26-4302-32P (32-pin) mates with J908; 26-4305-24P (24-pin) mates with J913; and 26-4306-24P (24-pin) mates with J911. The function of the cable is to provide an interconnection between the AN/ARM-155 Indicator Test Set and the AN/ARN-105 TACAN for test purposes.

3.5.7.2.10 Test bench harness AN/ARM-84 (8'2"). This Harness Assembly (Part No. E-35533) shall be 98 inches in length consisting of an adapter and five attached cables. The AN/ARN-52 TACAN fits into the adapter. The four cables on one end are Amphenol connectors and mate with C-8963/ARM-155 as follows: 26-4306-24P (24-pin) mates with J911, 26-4302-32P (32-pin) mates with J908, 26-4304-24P (24-pin) mates with J912, and 26-4303-24P (24-pin) mates with J910. ITT Cannon connector (on the other end) MS3126E16-26P mates with the AN/ARN-52(V) TACAN. The function of the harness is provided with an interconnection between the AN/ARM-155 Indicator Test Set and the AN/ARN-52(V) TACAN for test purposes. In addition, a blower is incorporated into the adapter assembly to provide cooling air for the TACAN.

3.5.7.2.11 Cable assembly W-3/ASM-85 (9'10"). This cable shall be 118 inches in length consisting of one connector on one end and six connectors on the other end. The six Amphenol connectors mate with C-8963/ARM-155 as follows: 26-4305-24P (24-pin) mates with J913; 26-4304-24P (24-pin) mates with J912; 26-4306-24P (24-pin) mates with J911; 26-4303-24P (24-pin) mates with J910; 26-4303-32P (32-pin) mates with J909; and 26-4302-32P (32-pin) mates with J908. The single connector at the other end of the cable shall be an Amphenol MS3106A36-10S (48-pin) which shall mate with J105 of the Distribution Box-Power Supply J-2022/ASM-85.

3.5.7.1.12 Test bench harness (North Island Part No. E36361) AN/ARN-86. This test harness is connected between the C-8963/ARM-155 connectors J908, J910, J911, J912, and J913 and the AN/ARN-86 TACAN for testing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

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1. Preproduction Tests

2. Initial Production Tests

3. Acceptance Tests

4. Life Test

4.2 Preproduction tests. Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction test data. The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts.

4.2.2 Scope of tests. Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction test shall include environmental and interference tests in accordance with the requirements of MIL-T-5422 to the limits specified in MIL-T-21200, except as modified herein (see 3.3.10).

4.2.2.1 Maintainability assurance test. A maintainability assurance test shall be conducted in accordance with MIL-STD-471 to demonstrate the requirements 3.3.7. The test details, shall include as a minimum, task selection, test method, demonstration support material, demonstration conditions, demonstration test team, test schedule and test reports, and shall be a part of the test procedure, in accordance with 4.6, to be submitted to the procuring activity for approval.

4.2.3 Preproduction approval. Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

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4.2.4 Production equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause of rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial production tests. One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests (see 4.4.1). The preproduction sample shall not be selected for this test.

4.3.1 Scope of tests. This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory material. In addition to the complete equipment submitted for initial production tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial production sample approval. Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of initial production test sample. On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and test shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such

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reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of materials during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

1. Individual Tests
2. Sampling Tests
3. Reliability Assurance Tests
4. Special Tests

4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

1. Examination of Product
2. Operational Test
3. Manufacturing Run-in Test

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the materials and workmanship requirements have been met.

4.4.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing run-in test. Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of 3g.

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The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failure occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling tests. Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0*
Next 50	1
Next 75	1
Next 100	1

*NOTE: If by contractual action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of tests. As a minimum each equipment selected by sampling tests shall be subjected to the following tests:

1. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specifications requirements have been met.

2. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should

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be based on results of the preproduction, initial production, individual and special tests.

3. Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours. Each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

Should repetitive failures occur, corrective action shall be required to eliminate the defect. Special tests (4.4.4) shall be initiated to satisfy the procuring activity that the defect has been corrected.

4.4.3 Reliability assurance tests. Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests shall be conducted as required by the Production Qualification Phase and the Production Acceptance Sampling Phase.

4.4.3.1 Production qualification phase. Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase Test Plan IV, Test Level E shall be used except that -54° C shall be changed to -20° C; the "on" time of any single temperature cycle shall be 20 hours; the off time of any single cycle shall be 4 hours; and the rate of change of the thermal medium shall average not less than 2° C/minimum.

4.4.3.2 Reliability production acceptance sampling phase tests. Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Plan IV, Test Level E shall be used except that -54° C shall be changed to -20° C; the "on" time of any single temperature cycle shall be 20 hours; the off time of any single cycle shall be 4 hours; and the rate of change of the thermal medium shall average not less than 2° C/minimum.

4.4.3.2.1 Lot size for sampling phase. The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be considered one lot unless otherwise specified in the contract. The first four available production units from the lot shall constitute the sample to be tested unless otherwise specified in the contract. The sample shall be tested until an accept or reject decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts and the equipment quality control.

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4.4.4 Special tests. Special tests shall be conducted on a quantity of equipment for the purpose of checking the effect of any design or materials change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

1. On an early equipment after an engineering or material change.

2. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

1. Determine the cause of failure.

2. Determine if the failure is an isolated case or design defect.

3. Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

4. Where practical, include a test in the Individual Tests of 4.4.1 to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the Life Test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the Individual Tests. The Life Test shall be performed under the conditions specified in 4.5.1. The Life Test sample shall be selected by the government inspector. Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests or Special Tests may be selected for Life Tests. When Reliability Tests are conducted, the Life Test may be omitted if, during the reliability tests, a quantity of equipments

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equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 50	1
Next 100	1
	1 for each additional 300 or fraction thereof

4.5.1 Test conditions. The Life Test shall be conducted under the following simulated service conditions:

Temperature	Room Ambient (30° ±10° C)
Altitude	Normal ground (0 - 5000 ft)
Humidity	Room Ambient
Input Voltage	115 ±10% (at lowest applicable frequency)
Vibration	None

4.5.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance check. At approximately 8-hour intervals during the test, a limited Performance Check shall be made. The Performance Check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

1. Part type number
2. The circuit reference symbol number

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3. The part function

4. Name of manufacturer

5. Nature of the failure

6. The number of hours which the part operated prior to failure.

4.5.4.1 Failure report. In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of life test samples. An equipment which has been subjected to the life test shall be reconditioned as follows:

1. On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

2. After reworking the contractor shall resubmit the equipment for acceptance.

4.6 Test procedures. The procedures used for the conducting Preproduction Tests, Acceptance Tests and Life Tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity for the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of tested equipment. Equipment which has been subjected to Acceptance and Life Tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission testing. No item, part or complete equipments shall be submitted by the contractor until it has been previously tested and inspected

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by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the equipment shall be preserved packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. The Test Set specified by this document is intended for use in the maintenance, development, and production of airborne TACAN receiver-transmitters and related systems. This Test Set shall be stackable with the units of Test Set, Radio, AN/ARM-22 and Test Set, Radio, AN/ARM-156, and shall provide mounting, input power control, precision and aircraft analog indicators and metering facilities, digital indicators, two-way data link indicators, data link enabling control, for use as required in the maintenance of navigation, identity, Inverse TACAN, Air-to-Air and Air to Ground Data Link and Stationkeeping functions as provided in the AN/ARN-21, AN/ARN-52, AN/ARN-84, AN/ARN-86, AN/ARN-105, AN/ASQ-19, AN/ASQ-56, AN/ASQ-57, AN/ASQ-58, AN/ASQ-88, or equivalent airborne TACAN systems.

6.2 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accomplished by complete supporting information.

6.3 Non-repairable subassemblies. As a general rule, Non-Repairable Subassemblies shall be encapsulated or hermetically sealed. The number of connections internal to the subassembly shall be held to a minimum. Detail parts tolerances and ratings shall be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage

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power supplies), the Non-Repairable Subassembly shall evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

1. Contract - The contract shall have precedence over any specification.

2. This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

3. Referenced specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

1. Title, number, and date of this specification.

2. Selection of applicable levels of packaging and packing (see 5.1).

6.7 Associated equipment. The Test Set, Indicator AN/ARM-155 shall operate with the following:

1. Test Set, Radio, AN/ARM-156 or Test Set, Radio, AN/ARM-22.

Preparing activity:
Navy - AS
(Project No. 5826-N127)

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Naval Air Systems Command
Washington, D.C. 20361

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-81988(AS) Test Set, Indicator AN/ARM-155

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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